The Duel at Saint Denis

Time limit per test: 2 seconds
Memory limit per test: 1 gb

The town of Saint Denis is preparing for a final evening of order and unrest.

Arthur Morgan observes a long street where **N** outlaws are standing in a straight line. Each outlaw is assigned a strength value, which determines how dangerous they are in a confrontation.

Arthur is not interested in full-scale chaos. Instead, he is studying controlled duels to understand how strength evolves under distance and positioning.

He will choose exactly two outlaws positioned at indices **i** and **j**, where **i** < **j**, and simulate a controlled duel scenario between them.

The outcome of the duel is not determined only by their strength. Distance between them also plays a critical role, as time spent preparing, walking, and observing the opponent affects performance.

If Arthur selects outlaws at positions **i** and **j**, the weaker of the two outlaws gains advantage from the distance between them. The duel value is defined as:

$$\min(a_i, a_j) + (\mathbf{j} - \mathbf{i})$$

Here, a_i and a_j are the strengths of the chosen outlaws, and **j - i** represents the distance factor between them.

Arthur wants to determine the maximum possible duel value among all valid pairs of outlaws.

Input:

The first line contains a single integer **t** representing the number of test cases.

Each test case consists of:

1) A single integer **n** ($2 <= \mathbf{n} <= 2 \times 10^5$) representing the number of outlaws.

2) A line containing **n** integers $a_1, a_2, ..., a_n$ ($1 <= a_i <= 10^{18}$) representing the strengths of the outlaws.

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5.

Output:

For each test case, output a single integer: the maximum possible duel value.

Example:

Input
3 5 4 2 7 1 9 4 5 5 5 5 3 10 1 20
Output
9 8 12

Explanation:

Test case 1

Choose outlaws at positions 3 and 5:

$$\min(7, 9) + (5 - 3) = 7 + 2 = 9$$

No other valid pair yields a higher duel value.

Test case 2

All outlaws have equal strength, so choosing positions 1 and 4 gives:

$$\min(5, 5) + (4 - 1) = 5 + 3 = 8$$

Test case 3

Choose outlaws at positions 1 and 3:

$$\min(10, 20) + (3 - 1) = 10 + 2 = 12$$

Notes:

Outlaws must be chosen from two different positions.

The first chosen position must appear before the second one.

Very strong outlaws might not always be optimal partners, and distance can compensate for lower strength.